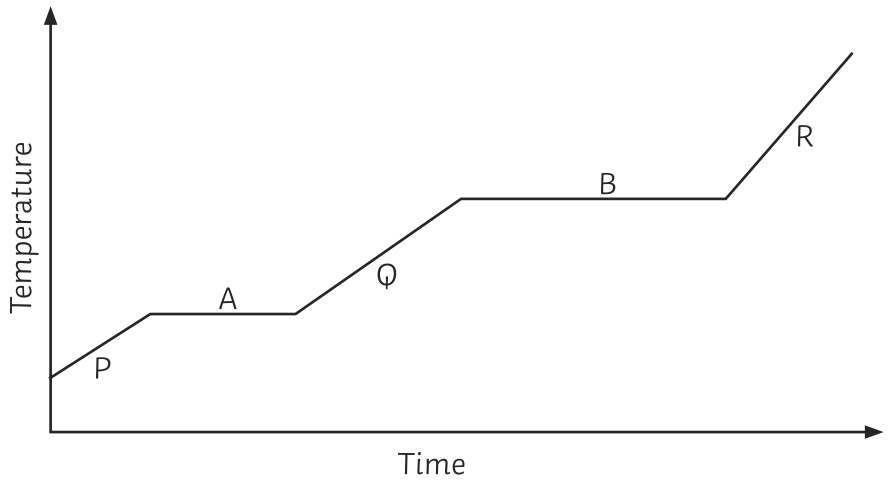
Name:	Class:	Date given:
		Date due in:

Changes of State **(Energetics)**



1. Does the graph show heating or cooling? Explain how you know. [2]

2. What state of matter is the substance in at point:

   P? _____

   Q? _____

   R? _____

3. What process is happening at:

   A? _____

   B? _____

4. Explain the shape of the graph, with respect to the change between gradient and flat parts and what they show us. [4]

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Use the graph to state whether the changes of state during heating are an exothermic or endothermic reaction? [1]

On the graph paper below, sketch a fully labelled graph to show a cooling curve. [5]



Learning Outcomes (tick if achieved)

Q1	I can identify a heating or cooling curve	
Q2,3	I can identify states of matter and changes of state on a graph	
Q4	I can explain the shape of a heating or cooling curve	
Q5	I can describe a reaction as endothermic or exothermic	
Q6	I can draw a heating or cooling curve	

Changes of State (Energetics) **Answers**

1. Does the graph show heating or cooling? Explain how you know. [2]

   It shows heating. I know because the temperature increase over time, solid melts over time, liquid evaporates/boils over time.

2. What state of matter is the substance in at point:

   P? **solid**

   Q? **liquid**

   R? **gas**

3. What process is happening at:

   A? **melting**

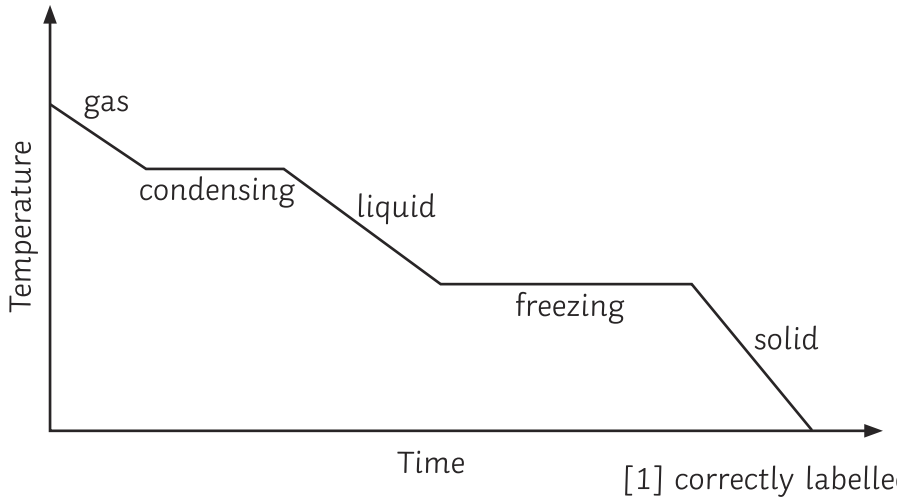
   B? **boiling/evaporating**

4. Explain the shape of the graph, with respect to the change between gradient and flat parts and what they show us. [4]

   Energy is supplied from heating
   Energy is used for raising temperature at points PQR
   Gradient indicates temperature increase
   Energy used for weakening forces / changing particle arrangement at points AB

5. Use the graph to state whether the changes of state during heating are an exothermic or endothermic reaction? [1]

   Endothermic reaction

6. 

   [1] correctly labelled axes
   [1] overall decline in graph
   [1] general shape with 3 gradient / 2 flat sections
   [1] states of matter correctly labelled
   [1] changing state processes correctly labelled